

ECO 202 — Practice Exam 3 — Version B

Statistics and Data Analysis for Economics | Princeton University

Fall 2026 | Classes 10–13 | 100 points

Name: _____

Student ID: _____

Timing. Target workload: 70 minutes (10 minutes multiple choice; 60 minutes written work). Follow announced start/end times within the 80-minute meeting. For practice, set a 70-minute timer.

Conditions. No books, notes, calculators, AI, or other electronics; University-authorized accommodations apply. Fractions, radicals, and equivalent exact expressions are acceptable. Show reasoning in written answers. Data are teaching examples unless identified otherwise.

Points. Questions 1–4: one answer A–D, 5 points each, no negative marking. Questions 5–8: 20 points each; subpart points are shown. Label any continuation clearly.

Multiple choice (20 points; about 10 minutes)

1. A discrete variable has $F_X(2) = 3/4$ and $\mathbb{P}(X < 2) = 1/4$. What is $\mathbb{P}(X = 2)$?
 - A. $1/4$.
 - B. $3/4$.
 - C. $1/2$.
 - D. 1.
2. For 16 independent observations with common mean μ and variance 9, which statement is exact without requiring a Normal population?
 - A. $\bar{X}_{16}$ has a Normal distribution.
 - B. $\text{SD}(\bar{X}_{16}) = 9/16$.
 - C. $\mathbb{E}[\bar{X}_{16}] = \mu$ and $\text{Var}(\bar{X}_{16}) = 9/16$.
 - D. $\bar{X}_{16} = \mu$.
3. Under independent, identically distributed sampling with finite positive variance, which distinction is correct?
 - A. The LLN concerns concentration around the mean; the CLT concerns the distribution of centered and scaled sampling error.
 - B. The LLN and CLT both say every observed error decreases when n increases.
 - C. The CLT states that the population becomes Normal.
 - D. The LLN supplies an exact Normal tail probability at every sample size.
4. Suppose $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, and $\text{Corr}(X, Y) = 1/2$. What is $\text{Var}(X - Y)$?
 - A. 13.
 - B. 12.
 - C. 19.
 - D. 7.

5. A nonlinear transformation can preserve dependence (20 points; about 15 minutes)

A discrete variable X has distribution

x	-1	0	1
$\mathbb{P}(X = x)$	$1/4$	$1/4$	$1/2$

Define $H = X^2$.

- (a) **(6 points)** Write the CDF $F_X(t)$ as a piecewise function. Calculate $\mathbb{E}[X]$ and $\text{Var}(X)$.
- (b) **(4 points)** Give the distribution of H and compute its expectation and variance. Express H as the indicator of an event involving X .
- (c) **(6 points)** Calculate $\mathbb{E}[XH]$, $\text{Cov}(X, H)$, and $\text{Var}(X + H)$.
- (d) **(4 points)** Use one event-probability comparison to show directly that X and H are dependent. Explain why treating their product expectation as the product of expectations would be incorrect.

6. Estimating a population maximum (20 points; about 15 minutes)

A population variable takes values 0, 2, and 4 with probabilities $1/2$, $1/4$, and $1/4$, respectively. Draw independently with replacement and let $M_2 = \max(X_1, X_2)$. The target is the largest possible population value, $m = 4$.

You may use $\text{MSE}(T; m) = \mathbb{E}[(T - m)^2] = \text{Var}(T) + (\mathbb{E}[T] - m)^2$.

- (a) **(6 points)** Find the complete sampling distribution of M_2 . Explain how the event $M_2 \leq 2$ helps determine the probabilities.
- (b) **(5 points)** Calculate $\mathbb{E}[M_2]$ and its bias as an estimator of m . Explain why the bias has this sign.
- (c) **(5 points)** Calculate $\text{MSE}(M_2; m)$ directly from the sampling distribution. What distinct notion does MSE capture compared with bias alone?
- (d) **(4 points)** For n independent draws, let M_n be their maximum. Derive an exact expression for $\mathbb{P}(M_n = 4)$ and explain why the original two-draw distribution cannot be retained when n changes.

7. A discrete count and its continuous approximation (20 points; about 15 minutes)

Suppose $H_1, \dots, H_{48}$ are independent Bernoulli indicators for a specified household characteristic, each with probability $p = 1/4$. Let $K = \sum_{i=1}^{48} H_i$ and $\hat{P} = K/48$.

Use $\Phi(z) = \mathbb{P}(Z \leq z)$ for standard Normal Z , with $\Phi(11/6) \approx 0.9666$ and $\Phi(5/3) \approx 0.9522$.

- (a) **(6 points)** Give the exact distribution family and parameters of K . Derive the means and standard errors of K and $\hat{P}$. Check the expected numbers of successes and failures.
- (b) **(6 points)** Use a continuity-corrected Normal approximation to calculate $\mathbb{P}(K \leq 17)$. Show the boundary and standardization. Give the equivalent event for $\hat{P}$.
- (c) **(4 points)** Extend the model to n indicators. Explain the different limiting behavior described by the LLN for $\hat{P}_n$ and by the CLT for $(\hat{P}_n - p)/\sqrt{p(1-p)/n}$.
- (d) **(4 points)** A report says, “Because the expected count is 12, exactly 12 households in this sample must have the characteristic.” Repair the claim and distinguish the expected count from one realized count.

8. Combining estimators and reporting uncertainty (20 points; about 15 minutes)

Under one sampling model, T_1 is unbiased for θ_1 , T_2 is unbiased for θ_2 , and

$$\text{Var}(T_1) = 4, \quad \text{Var}(T_2) = 9, \quad \text{Cov}(T_1, T_2) = 1.$$

You may use $\text{MSE}(T; \theta) = \text{Var}(T) + (\mathbb{E}[T] - \theta)^2$.

- (a) **(5 points)** Show whether $T_1 - 3T_2$ is unbiased for $\theta_1 - 3\theta_2$. Does your expectation argument require independence?
- (b) **(5 points)** Find the variance and MSE of $T_1 - 3T_2$ for that target. Explain why simply adding $4 + 9(9)$ would be incorrect.
- (c) **(5 points)** Is $T_1^2 + T_2$ unbiased for $\theta_1^2 + \theta_2$? Find its bias and give an unbiased correction. No variance calculation for this nonlinear estimator is required.
- (d) **(5 points)** In a separate independent sample of 25 invoices, the observed mean is 80 dollars and sample SD is 10 dollars. Calculate the estimated SE of the mean. An analyst says this establishes that the estimate missed the population mean by exactly that amount. Explain the error and what information is missing.